

SELVAGANESH V

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Artificial Intelligence & Machine Learning

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[GitHub](#) • [LinkedIn](#) • [Portfolio](#) • [Hugging Face](#)

CAREER OBJECTIVE

Artificial Intelligence and Machine Learning graduate (B.Tech, 2026) with hands-on experience building and deploying end-to-end AI applications: Retrieval-Augmented Generation with FAISS semantic search, multilingual NLP with Hugging Face Transformers, and Azure OpenAI vision. Skilled in shipping production-ready apps with FastAPI, Gradio, and Docker on cloud platforms.

SKILLS

Programming: Python

AI/ML: Machine Learning, Deep Learning, NLP, Hugging Face, Azure OpenAI

Frameworks: Gradio, FastAPI, Flask, Uvicorn

Cloud & Tools: Docker, Render, Supabase, Vercel, Netlify, N8N

APIs: OpenRouter, Groq, REST APIs

EDUCATION

B.Tech Artificial Intelligence & Machine Learning	2022–2026
Bannari Amman Institute of Technology, Sathyamangalam	CGPA: 7.51
Diploma in Electrical & Electronics Engineering	2020–2023
PSG Polytechnic College, Coimbatore	83.41%

PROJECTS

PDF-RAG Application [GitHub](#) | [Live Link](#)

- Built a Retrieval-Augmented Generation (RAG) system in **Python** that extracts and indexes PDF text using **PyMuPDF**.
- Generated **all-MiniLM-L6-v2** embeddings and ran top-k similarity search with **FAISS** for context retrieval.
- Produced grounded answers via an **OpenRouter LLM API** with **SHA-256 caching** for efficiency.
- Deployed an interactive **Gradio** app on **Hugging Face Spaces** supporting multi-PDF upload and real-time Q&A.

Emotion Detection in Text [GitHub](#) | [Live Link](#)

- Built a multilingual sentiment app in **Python** using the RoBERTa model **cardiffnlp/twitter-roberta-base-sentiment** (~70% TweetEval accuracy).
- Classified text into **Positive, Neutral, and Negative** sentiments with **PyTorch** inference over Hugging Face Transformers.
- Integrated **langdetect** and **Google Translator** to auto-detect any language and translate to English before prediction.
- Deployed an interactive **Gradio** web app on Hugging Face Spaces for real-time analysis.

Crop Identification System [GitHub](#) | [Live Link](#)

- Built a crop identification and farming-assistant web app in **Python** using the **Azure OpenAI** vision model for image-based recognition.
- Preprocessed uploads with **PIL** (resize, RGB, JPEG compression) and **base64** encoding for vision-API inference.
- Added **MD5-hash caching** and **session memory** to power a contextual farming-advisor chatbot on top of identification.
- Deployed an interactive **Gradio** (PWA-enabled) app with image upload and multi-turn Q&A.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified	2025	Certificate
Python for Scientific Computing – freeCodeCamp	2025	Certificate
Generative AI Workshop (NXT Wave, IIT Delhi alumnus)	2024	Certificate

LANGUAGES

Tamil (Read, Write, Speak) • English (Read, Write, Speak)